
**Information technology —
Radio frequency identification for item
management —**

**Part 7:
Parameters for active air interface
communications at 433 MHz**

*Technologies de l'information — Identification par radiofréquence
(RFID) pour la gestion d'objets —*

*Partie 7: Paramètres pour les communications actives d'une interface
d'air à 433 MHz*

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Published in Switzerland

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Foreword

ISO (the International Organization for Standardization) and IEC (the International Electrotechnical Commission) form the specialized system for worldwide standardization. National bodies that are members of ISO or IEC participate in the development of International Standards through technical committees established by the respective organization to deal with particular fields of technical activity. ISO and IEC technical committees collaborate in fields of mutual interest. Other international organizations, governmental and non-governmental, in liaison with ISO and IEC, also take part in the work. In the field of information technology, ISO and IEC have established a joint technical committee, ISO/IEC JTC 1.

International Standards are drafted in accordance with the rules given in the ISO/IEC Directives, Part 2.

The main task of the joint technical committee is to prepare International Standards. Draft International Standards adopted by the joint technical committee are circulated to national bodies for voting. Publication as an International Standard requires approval by at least 75 % of the national bodies casting a vote.

ISO/IEC 18000-7 was prepared by Technical Committee ISO/IEC JTC 1, *Information technology*, Subcommittee SC 31, *Automatic identification and data capture techniques*.

ISO/IEC 18000 consists of the following parts, under the general title *Information technology — Radio frequency identification for item management*:

- *Part 1: Reference architecture and definition of parameters to be standardized*
- *Part 2: Parameters for air interface communications below 135 kHz*
- *Part 3: Parameters for air interface communications at 13,56 MHz*
- *Part 4: Parameters for air interface communications at 2,45 GHz*
- *Part 6: Parameters for air interface communications at 860 MHz to 960 MHz*
- *Part 7: Parameters for active air interface communications at 433 MHz*

Introduction

This part of ISO/IEC 18000 is intended to address RFID devices operating in the 433 MHz frequency band, providing an air interface implementation for wireless, non-contact information system equipment for Item Management applications. Typical applications operate at ranges greater than one meter.

The RFID system includes a host system and RFID equipment (interrogator and tags). The host system runs an application program, which controls interfaces with the RFID equipment. The RFID equipment is composed of two principal components: tags and interrogators. The tag is intended for attachment to an item, which a user wishes to manage. It is capable of storing a tag ID number and other data regarding the tag or item and of communicating this information to the interrogator. The interrogator is a device, which communicates to tags in its RF communication range. The interrogator controls the protocol, reads information from the tag, directs the tag to store data in some cases, and ensures message delivery and validity. This system uses an active tag.

RFID systems defined by this part of ISO/IEC 18000 provide the following minimum features:

- Identify tag in range
- Read data
- Write data or handle read only systems gracefully
- Selection by group or address
- Graceful handling of multiple tags in the field of view
- Error detection

The International Organization for Standardization (ISO) and International Electrotechnical Commission (IEC) draw attention to the fact that it is claimed that compliance with this document may involve the use of a patent concerning radio-frequency identification technology given in sub-clause 6.2.

ISO and IEC take no position concerning the evidence, validity and scope of this patent right.

The holder of this patent right has assured the ISO and IEC that they are willing to negotiate licenses under reasonable and non-discriminatory terms and conditions with applicants throughout the world. In this respect, the statement of the holder of this patent right is registered with ISO and IEC. Information may be obtained from the following companies.

Contact details	Patent holder	Patent title	Patent number	Affected subclause in this part of ISO/IEC 18000
Ravi Rajapaksi, Chief Technology Officer, Savi Technology, Inc., 615 Tasman Dr., Sunnyvale, CA 94089 USA	Savi Technology	Communication System for Communicating with Tags	US 5640151	6.2.6
	Savi Technology	Communication System for Communicating with Tags	US 5686902	6.2.6
	Savi Technology	Method and Apparatus for Radio Identification and Tracking	EP 0467036	6.2.6
Rob Sokohl, Sterne, Kessler, Goldstein & Fox P.L.L.C., 1100 New York Avenue NW, Washington, DC 20005-3934 USA	Matrics Technology	System and method for electronic inventory	US 6002344	6.2

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights other than those identified above. ISO or IEC shall not be held responsible for identifying any or all such patent rights.

Information technology — Radio frequency identification for item management —

Part 7: Parameters for active air interface communications at 433 MHz

1 Scope

This part of ISO/IEC 18000 defines the air interface for radio frequency identification (RFID) devices operating as an active RF Tag in the 433 MHz band used in item management applications. The purpose of this part of ISO/IEC 18000 is to provide a common technical specification for RFID devices that may be used by ISO committees developing RFID application standards. This part of ISO/IEC 18000 is intended to allow for compatibility and to encourage inter-operability of products for the growing RFID market in the international marketplace. This part of ISO/IEC 18000 defines the forward and return link parameters for technical attributes including, but not limited to, operating frequency, operating channel accuracy, occupied channel bandwidth, maximum power, spurious emissions, modulation, duty cycle, data coding, bit rate, bit rate accuracy, bit transmission order, and where appropriate operating channels, frequency hop rate, hop sequence, spreading sequence, and chip rate. This part of ISO/IEC 18000 further defines the communications protocol used in the air interface.

2 Conformance

The rules for RFID device conformity evaluation will be given in a future Technical Report (ISO/IEC TR 18047-7).

3 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 15963, *Information technology — Radio frequency identification for item management — Unique identification for RF tags*

ISO/IEC 19762-1, *Information technology — Automatic identification and data capture (AIDC) techniques — Harmonized vocabulary — Part 1: General terms relating to AIDC*¹⁾

ISO/IEC 19762-3, *Information technology — Automatic identification and data capture (AIDC) techniques — Harmonized vocabulary — Part 3: Radio frequency identification (RFID)*¹⁾

1) To be published.

4 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO/IEC 19762-1 and ISO/IEC 19762-3 apply.

5 Symbols and abbreviated terms

For the purposes of this document, the symbols and abbreviated terms given in ISO/IEC 19762-1 and ISO/IEC 19762-3 apply.

6 433,92 MHz active narrowband specification

6.1 Physical layer

The RF communication link between interrogator and tag utilizes narrow band UHF frequency with the following characteristics:

Carrier Frequency	433,92 MHz $\pm$ 20 ppm
Modulation Type	FSK
Frequency deviation	$\pm$ 50 kHz
Symbol LOW	fc +50 kHz
Symbol HIGH	fc –50 kHz
Modulation rate	27,7 kHz
Wake up Signal	30 kHz

The Wake up signal is transmitted by interrogator for minimum of 2,5 seconds to wake up all tags within communication range. The wake up signal is a 30 kHz sub-carrier tone for 2,5 to 2,7 seconds. Upon detection of the Wake up signal all tags will enter into Ready state awaiting command from the interrogator.

The communication between interrogator and tag is of Master-Slave type, where the interrogator always initiates communications and then listens for response from a tag. Multiple response transmissions from the tags are controlled by collection algorithm described in section “Tag Collection and Collision Arbitration”.

6.2 Data Link layer

6.2.1 General

Data between interrogator and tag is transmitted in packet format. A packet is comprised of a preamble, data bytes and a final logic low period. The end of preamble and beginning of the first data byte is indicated by the last two pulses of the preamble. The same two pulses of the preamble also indicate the originator of the data packet. Data bytes are sent in Manchester code format. Transmission order is most significant byte first; within a byte, the order is least significant bit first. Figure 1 illustrates the data communication timing of the preamble and the first byte of a packet.

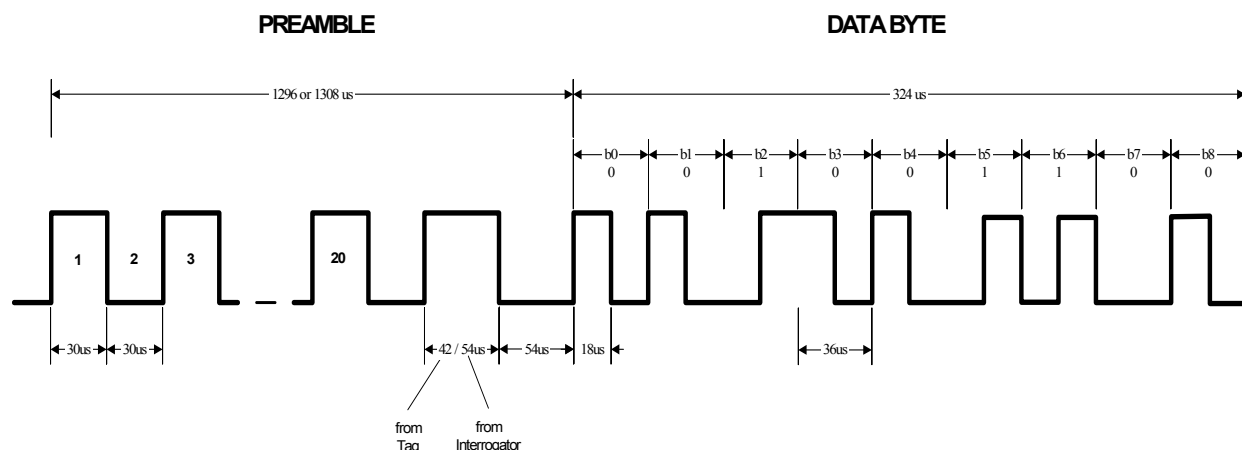


Figure 1 — Data communication timing

6.2.2 Preamble

The preamble is comprised of twenty pulses of 60 μ s period, 30 μ s high and 30 μ s low, followed by a final sync pulse which identifies the communication direction: 42 μ s high 54 μ s low (Tag to Interrogator); 54 μ s high 54 μ s low (Interrogator to Tag).

6.2.3 Data byte format

Data bytes are in Manchester code format, comprised of 8 data bits and one stop bit. The bit period is 36 μ s, the total byte period is 324 μ s. A falling edge in the center of the bit-time indicates a 0 bit, a rising edge indicates a 1 bit. The stop bit is coded as a zero bit.

6.2.4 CRC bytes

A CRC checksum is calculated as a 16-bit value, initialized with all ones ('FFFF'), over all data bytes (excluding preamble) according to the CCITT polynomial ($x^{16} + x^{12} + x^5 + 1$). The CRC is appended to the data as two bytes. Reference: "The CCITT Red Book", Volume VIII, International Telecommunications Union, Geneva, 1986. Recommendation V.41, "Code-Independent Error Control System."

6.2.5 Packet end period

A final period of 36 μ s of continuous logic low is transmitted for each packet after the CRC bytes.

6.2.6 Interrogator to tag message format

Tags shall recognize the following message format:

Table 1 — Interrogator to tag message format

Optional fields depending on the Command Type					Optional field depending on the Command Type		
MSB					LSB		
Command Prefix	Command Type	Owner ID	Tag ID	Interrogator ID	Command Code	Parameters	CRC
1 byte (‘31’)	1 byte (8 bits)	3 bytes	4 bytes	2 bytes	1 byte	N bytes	2 bytes

The Owner ID, Tag ID and Parameters fields are present as required by the Command Type and Command code described in the following sections.

6.2.6.1 Command type

Table 2 — Command type field

Bit							
7	6	5	4	3	2	1	0
Reserved	Reserved	Reserved	Reserved	Reserved	1	0= Broadcast (Tag Id not present) 1= Point to Point (Tag ID present)	0= Owner ID not present 1= Owner ID present

The Command Type field is used to indicate the presents of Tag ID and Owner ID fields within the current data packet. If Interrogator wishes to address a single tag by specifying its Tag ID, Bit 1 of the Command Type field needs to be set, indicating point-to-point communication. In the case Interrogator wants to address all the tags within its RF communication range, Bit 1 of the Command Type field needs to be cleared to indicate broadcast message. A broadcast message does not use Tag ID field and is omitted from the data packet.

The Bit 0 of the Command Type field indicates whether Owner ID is included in Interrogator to Tag message. If the Owner ID is included in the message all tags within Interrogator's RF communication range that belongs to the same Owner ID will respond back. All other tags shall ignore such message.

6.2.6.2 Owner ID

The Owner ID field allows Interrogator to communicate with only those tags that belong to a specific owner ID group. The Owner ID can be arbitrarily programmed and subsequently changed within each tag's non-volatile memory. If the tag is not programmed with Owner ID or its value is set to zero tag shall respond to any Interrogator command that does not include Owner ID in the command message.

6.2.6.3 Tag ID

The Tag ID is a 32-bit integer number that is uniquely assigned to each individual tag during manufacturing. This number cannot be changed and is read only. The Tag ID number has no structure and does not contain any information besides uniquely identifying a tag. The Tag ID cannot be reused. Issuance of the Tag IDs may be managed and administered by UCC/EAN, ISO 14816, ISO 6346, or some comparable international organization on behalf of the associated manufacture and is regulated by provision of agreement. ISO/IEC 15963 describes the structure of the Tag ID.

6.2.6.4 Interrogator ID

The Interrogator ID is 16 bits integer number programmed into the Interrogator non-volatile memory. The Interrogator ID can be changed without any restrictions and is used to efficiently route tag responses through Interrogators network. The Interrogators that receive a tag message not addressed to them shall not pass the message to the system.

6.2.6.5 Command codes

The Command codes and their function as a Read and / or Write command are summarized below. The least significant 7 bits of a command identify its base function, the eighth (MS) bit is set '0' for a Read function and '1' for a Write function. Codes not identified are reserved.

Table 3 — Command codes

Command code (R / W)	Command name	Command type	Description
'10' / NA	Collection	Broadcast	Collect all Tag IDs within interrogator RF communication range
'11' / NA	Collection with Data	Broadcast	Collect all Tag IDs including specified data from tag's non-volatile memory
'14' / NA	Collection with User ID	Broadcast	Collect all Tag IDs including tag's User ID
NA / '15'	Sleep	Point to Point	Put Tag to sleep
'01' / NA	Status	Point to Point	Retrieve Tag status
'07' / '87'	User ID length	Point to Point	Sets length of the User ID (in bytes, 1 – 16)
'13' / '93'	User ID	Point to Point	Sets User assigned ID (1 – 16 bytes)
'09' / '89'	Owner ID	Point to Point	Set Owner ID (3 bytes)
'0C' / NA	Firmware revision	Point to Point	Set by manufacturer
'0E' / NA	Model Number	Point to Point	Set by manufacturer
'60' / 'E0'	Read/Write Memory	Point to Point	Memory data
NA / '95'	Set Password	Point to Point	Sets Tag Password (4 bytes long)
'17' / '97'	Set Password Protect	Point to Point	Sets and Clears Tag Secure bit which enables or disables password protected access to the tag
NA/'96'	Unlock	Point to Point	Unlocks password protected tag

The Command Type column indicates whether the command is broadcast (does not include Tag ID in the message) or point-to-point (includes Tag ID in the message).

6.2.7 Tag to interrogator message format

The Tag to Interrogator message has two different formats depending on the type of message being transmitted to the Interrogator.

There are two possible formats:

- Broadcast response message format
- Point-to-Point response message format

6.2.7.1 Broadcast response message format

This message format is used in response to all Interrogator broadcast commands, received by Interrogator within its communication range.

The broadcast commands are used to collect Tag IDs, User IDs or short blocks of data from the selected group of tags (all tags, particular tag type, etc.) using batch collection algorithm. See Collection Command section for more details.

The User ID field is optional and its use is controlled by the *Command Type* field bit '0' as specified in *Interrogator to Tag message format*.

Table 4 — Broadcast response message format

Tag Status	Message Length	Int ID	Tag ID	Owner ID	User ID	Data	CRC
2 bytes	1 byte	2 bytes	4 bytes	3 bytes	0-16 bytes	0 – N bytes	2 bytes

Tag Status: Indicates various conditions such as response format, tag type, alert flag. See clause 6.2.5.2.3, **Tag Status**, for more details.

Msg. Length: Message length in bytes including CRC byte codes

Int ID: ID of Interrogator. Integer value from 1 to 65535

Tag ID: Unique Tag ID preset during manufacturing

Owner ID: Unique ID assigned to corporation.

User ID: User ID is optional field set by the user

Data: Data field depends on the type of collection command being used to collect the tags. This data field can include User ID or specific data block from tag's memory.

CRC: CCITT code check bytes

6.2.7.2 Point-to-point response message format

This message format is returned to the Interrogator as a response to all point-to-point commands, which require Tag ID in order to access particular tag (these include all commands except Collection commands).

Table 5 — Point-to-point response message format

Tag Status	Message Length	Int ID	Tag ID	Command Code	Parameters	CRC
2 bytes	1 byte	2 bytes	4 bytes	1 byte	N bytes	2 bytes

Tag Status: Indicates various conditions such as response format, tag type, alert flag. See clause 6.2.5.2.3, **Tag Status**, for more details.

Msg. Length: Message length in bytes including CRC byte codes

Int ID: ID of Interrogator, Integer value from 1 to 65535

Tag ID: Unique Tag ID preset during manufacturing

Command Code: This field contains received Command code

Parameters: This field contains parameters defined by Command code

CRC: CCITT code check bytes

6.2.7.3 Tag status

The Tag Status field, which is included in all Tag to Interrogator messages, consists of the following information:

Table 6 — Tag status field format

Bit							
15	14	13	12	11	10	9	8
Mode field				Reserved	Reserved	Reserved	Ack 1 = NAK 0 = ACK

Bit							
7	6	5	4	3	2	1	0
Reserved		Tag type				User ID 1 = Used 0 = Not Used	Battery 1 = low 0 = good

NOTE Reserved fields are set to a value of "0"

Mode field indicates response data format from the tag (Broadcast command, Point-to-Point command)

Table 7 — Tag status field format

Mode field	Mode format code (bit15 – 12)
Broadcast Command	0000
Point to Point Command	0010

Acknowledgment bit when set to '0' indicates a valid command (CRC ok and all fields valid) has been received from the Interrogator.

Tag type indicates tag capabilities or special features (memory size, sensor input, etc.). The code for Tags conforming to this part of ISO/IEC 18000 is '010'. All other codes are reserved.

User ID when set indicates that User ID is being included in the Collection response messages. If used only Tag ID is included in the Tag to Interrogator message.

Battery status bit when set indicates that ~80 % of the tag battery life is being reached.

6.2.7.4 Command codes

6.2.7.4.1 Collection

Table 8 — Collection command format

Command Code	Windows size	Reserved
'10'	2 bytes	1 bytes

See clause 6.2.6 for further information on “window size.”

Collection operation is used to interrogate and collect all tags. Reserved byte is always set to '00'.

6.2.7.4.2 Collection with data

Table 9 — Collection with data command format

Command Code	Windows size	Start Address (M)	Number of Data bytes (N)
'11'	2 bytes	3 bytes	1 byte

Where:

- **N** is integer number from 1 to 32
- **M** is address from 0 to maximum available tag memory

The Collection with Data operation is used to collect all tags and the contents of the their memory block specified through Start Address and Number of Data bytes parameters.

6.2.7.4.3 Collection with user ID

Table 10 — Collection with user ID command format

Command Code	Windows size
'14'	2 bytes

The Collection with User ID command is used to collect all tags with their User ID.

6.2.7.4.4 Sleep

Write

Table 11 — Sleep command format (Write)

Command code
'15'

The Sleep command directs a specific Tag to enter the Sleep mode. The Tag will not respond to this command nor to any subsequent command until the Tag is awoken again by Wake Up signal.

6.2.7.4.5 Read tag status

Read

Table 12 — Read tag status command format (read)

Command code
'01'

Read Response. Status byte is defined in 6.2.5.2.3.

Table 13 — Read tag status command format (read response)

Command code	Status
'01'	1 byte

6.2.7.4.6 Security — Set password

Write

Table 14 — Set password command format (write)

Command code	Password
'95'	4 bytes

Write Response

Table 15 — Set password command format (write response)

Command code
'95'

This command sets the tag's password. This command requires tag first to be unlocked before it can be accessed.

Initial value of the tag's password is to 'FFFFFFFF'.

6.2.7.4.7 Security — Set password protect

Write

Table 16 — Set password protect command format (write)

Command code	Secure
'97'	1 byte

- If Secure byte is set to '01', tag becomes password protected
- If Secure byte is set to '00', tag password protection condition is cleared

Write Response

Table 17 — Set password protect command format (write response)

Command code
'97'

This command sets or clears secure bit inside the tag. To clear (set to zero) secure bit, tag has to be first unlocked with proper password. If secure bit is set to '01', access to all point to point commands are password protected. Broadcast commands are not affected by the status of secure bit.

Initially secure bit is set to zero.

6.2.7.4.8 Security — Unlock

Write

Table 18 — Unlock command format (write)

Command code	Password
'96'	4 bytes

Write Response

Table 19 — Unlock command format (write response)

Command code
'96'

This command “unlocks” the access to the tag. If password ID matches tag's password tag shall respond to consecutive commands received from interrogator. If the password does not match, the tag shall not respond any point-to-point command. Tag shall remain in unlock state until it receives Sleep command or 30 seconds expires after unlock command was received.

6.2.7.4.9 Privacy — User ID length

Read

Table 20 — Privacy user ID length command format (read)

Command code
'07'

Read Response

Table 21 — Privacy user ID length command format (read response)

Command code	User ID Length
'07'	8 bits

Write

Table 22 — Privacy user ID length command format (write)

Command code	User ID Length
'87'	8 bits

Write response

Table 23 — Privacy user ID length command format (write response)

Command code
'87'

The User ID length determines the length of User ID.

6.2.7.4.10 Privacy — User ID

Read

Table 24 — User ID command format (read)

Command code
'13'

Read Response

Table 25 — User ID command format (read response)

Command code	User ID
'13'	0 – 16 bytes

Write

Table 26 — User ID command format (write)

Command code	User ID
'93'	1 – 16 bytes

Write response

Table 27 — User ID command format (write response)

Command code
'93'

The User ID has variable length (in 8 bit increments) and allows user to program its own user assigned asset ID. The length is defined through User ID Length field. User ID can be changed at any time and its interpretation is user specific.

6.2.7.4.11 Privacy — Owner ID

Read

Table 28 — Owner ID command format (read)

Command code
'09'

Read Response

Table 29 — Owner ID command format (read response)

Command code	Owner ID
'09'	3 bytes

Write

Table 30 — Owner ID command format (write)

Command code	Owner ID
'89'	3 bytes

Write Response

Table 31 — Owner ID command format (write response)

Command code
'89'

Owner ID provides additional layer of privacy defined by the owner of the Tag. If the Owner ID field is present in the Interrogator's command and the included ID does not match the Tag's Owner ID, the Tag will not respond to the command. If the tag's Owner ID is set to zero tag shall respond to any Owner ID specified by Interrogator.

6.2.7.5 Tag model and version

Following two commands are not mandatory for compliance with this part of ISO/IEC 18000.

6.2.7.5.1 Firmware version

Read

Table 32 — Firmware version command format (read)

Command code
'0C'

Read Response

Table 33 — Firmware version command format (read response)

Command code	Firmware version
'0C'	1 bytes

The Firmware Version indicates tag firmware version

6.2.7.5.2 Model number

Read

Table 34 — Model number command format (read)

Command code
'0E'

Read Response

Table 35 — Model number command format (read response)

Command code	Model number
'0E'	2 bytes

The Model Number indicates tag model number.

6.2.7.6 Read and write memory

6.2.7.6.1 Write memory

Write

Table 36 — Write memory command format (write)

Command Code	Number of bytes (N)	Start Address (M)	Data
'0E'	1 byte	3 bytes	N bytes

Response

Table 37 — Write memory command format (write response)

Command Code
'0E'

Where:

- N is integer number from 1 to 46
- M is address from 0 to maximum available tag memory

6.2.7.6.2 Read memory

Read

Table 38 — Read memory command format (read)

Command Code	Number of bytes (N)	Start Address (M)
'60'	1 byte	3 bytes

Response

Table 39 — Read memory command format (read response)

Command Code	Number of bytes (N)	Data
'60'	1 byte	N bytes

Where:

- **N** is integer number from 1 to 46
- **M** is address from 0 to maximum available tag memory

6.2.8 Tag collection and collision arbitration

The purpose of the collision arbitration sequence during tag collection is to perform an efficient and orderly collection of the tags placed within the interrogator communication range and to receive information on the tag capabilities and data contents in a single sequence. The information that the tag returns is specified by flags set in the command from the interrogator.

The interrogator is the master of the communication with one or multiple tags.

General explanation

The collision arbitration uses a mechanism, which allocates tag transmissions into slots within specified collection round (or so called window size). The minimum windows size is set to be 57,3 ms. A collection round consists of a number of slots. Each slot is long enough for the interrogator to receive a tag response. The actual duration of a slot is determined by the interrogator collection command type and is a function of the tag transmission time.

When the tags placed within the interrogator RF communication range receive “wake-up” signal that is broadcast by the interrogator, they will move into the Ready state.

The interrogator initiates a tag collection process by sending a Collection command. Tags receiving a Collection command randomly select a slot in which to respond, but the tags do not immediately start transmitting. The number of slots in a current collection round is determined by the interrogator command type. Each Collection command requires specific type and amount of data to be transmitted by the tag within single slot time. The initial window size is fixed and set to 57,3 ms. During the subsequent collision arbitration process, the interrogator dynamically chooses an optimum window size for the next collection round based on the number of collisions in the round. The number of collisions is a function of the number of tags present within the interrogator communication range that participate in the current collection round.

On receiving a Collection command, tags select a slot in which to respond. The selection is determined by a pseudo-random number generator. When a tag selects a slot_number it will wait for a pseudo-random time delay equal to a time of slot_number multiplied by slot_delay before it responds. The number of slots is determined by the current window size, indicated through the interrogator collection command type and a tag transmission time.

After the interrogator has sent the Collection command there are three possible outcomes:

- a) The interrogator does not receive a response because either no tag has selected current slot or interrogator did not detect a tag response. The interrogator then terminates current collection round.

- b) The interrogator detects a collision between two or more tag responses. Collisions may be detected either as contention from the multiple transmissions or by detecting an invalid CRC. The interrogator records collision and continue “listening” for a new tag in the subsequent slot.
- c) The interrogator receives a tag response without error, i.e. with a valid CRC. The interrogator records the tag data and continues to listen for a new tag in the subsequent slot.

The collection round continues until all slots within the round have been explored.

When the Collection round is completed the interrogator starts transmitting Sleep command to all tags collected during the previous Collection round. The tags that receive Sleep command move to “sleep” mode and will not participate in collection in the subsequent collection rounds.

The interrogator immediately starts the next collection round by transmitting the Collection command.

This process continues until no more tags are being detected during three subsequent collection rounds.

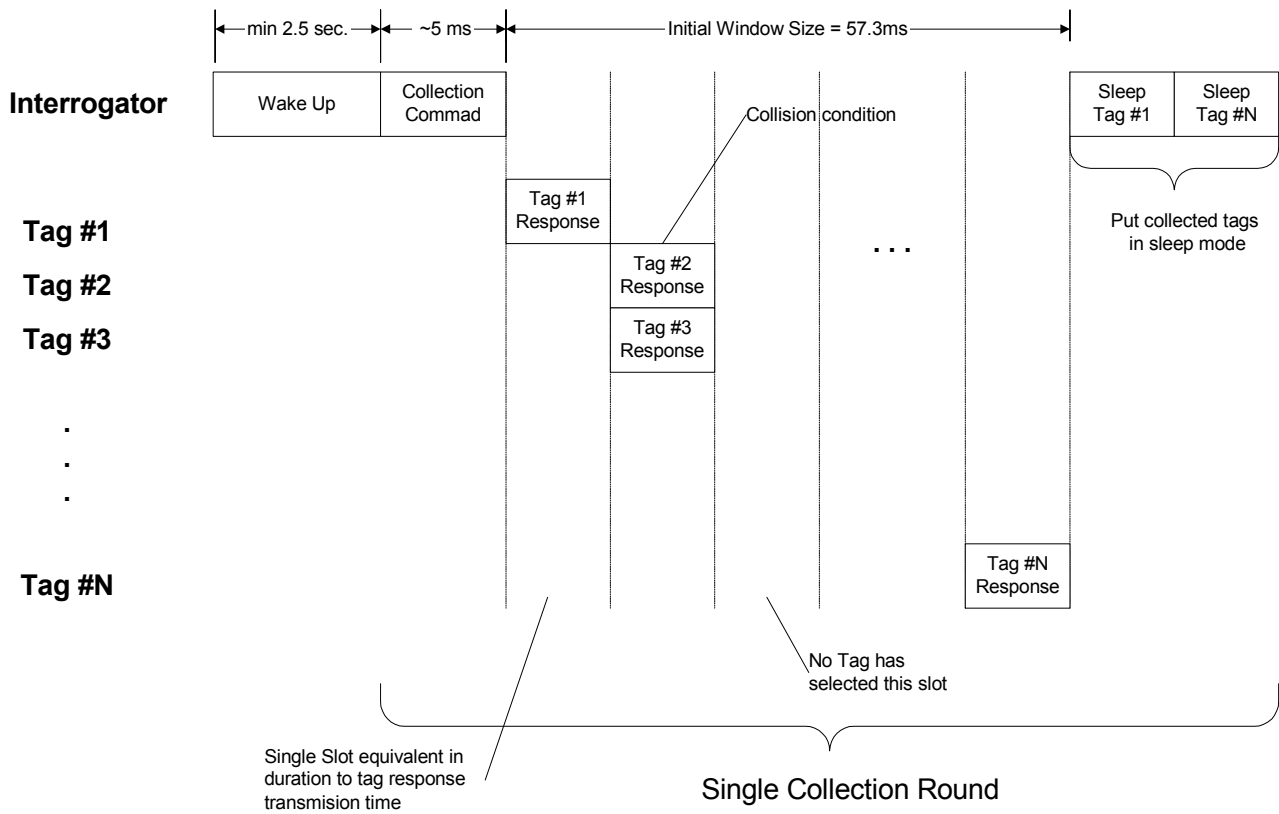


Figure 2 — Collection sequence and timing

6.3 Physical and Media Access Control (MAC) parameters

6.3.1 Interrogator to tag link

Table 40 — Interrogator to tag link parameters

Ref.	Parameter	Value
Int:1	Operating Frequency range	433,92 MHz
Int:1a	Default Operating Frequency	433,92 MHz
Int:1b	Operating Channels	Not applicable in this mode
Int:1c	Operating Frequency Accuracy	+/-20 ppm
Int:1d	Frequency Hop Rate	Not applicable in this mode
Int:1e	Frequency Hop Sequence	Not applicable in this mode
Int:2	Occupied Channel Bandwidth	500 kHz
Int:2a	Minimum Receiver Bandwidth	500 kHz
Int:3	Interrogator Transmit Maximum EIRP	5,6 dBm (peak power) or as allowed by local regulations
Int:4	Interrogator Transmit Spurious Emissions	
Int:4a	Interrogator Transmit Spurious Emissions, In-Band	Not applicable in this mode
Int:4b	Interrogator Transmit Spurious Emissions, Out of Band	The interrogator shall transmit in conformance with spurious emissions requirements defined by the country's regulatory authority within which the system is operated.
Int:5	Interrogator Transmitter Spectrum Mask	Not applicable in this mode
Int:6	Timing	
Int:6a	Transmit to Receive Turn Around Time	50 microseconds
Int:6b	Receive to Transmit Turn Around Time	50 microseconds
Int:6c	Interrogator Transmit Power On Ramp	1 ms
Int:6d	Interrogator Transmit Power Down Ramp	1 ms
Int:7	Modulation	Frequency Shift Keying (FSK)
Int:7a	Spreading Sequence	Not applicable in this mode
Int:7b	Chip Rate	Not applicable in this mode
Int:7c	Chip Rate Accuracy	Not applicable in this mode
Int:7d	Modulation Index	Not applicable in this mode
Int:7e	Duty Cycle	Not applicable in this mode
Int:7f	FM Modulation	± 35 kHz
Int:8	Data Coding	Manchester, 36 μ s bit period. Logic one: 18 μ s low followed by 18 μ s high, logic zero: 18 μ s high followed by 18 μ s low
Int:9	Bit Rate	27,7 kbit/s
Int:9a	Bit Rate Accuracy	200 ppm
Int:10	Interrogator Transmit Modulation Accuracy	Not applicable in this mode

Table 40 (continued)

Ref.	Parameter	Value
Int:11	Preamble	
Int:11a	Preamble Length	Twenty one (21) bits
Int:11b	Preamble Waveform	Square wave as defined in Int:11c
Int:11c	Bit Sync Sequence	20 cycles of 30 μ s high, 30 μ s low, followed by one cycle 54 μ s high, 54 μ s low
Int:11d	Frame Sync Sequence	20 cycles of 30 μ s high, 30 μ s low, followed by one cycle 54 μ s high, 54 μ s low
Int:12	Scrambling	Not applicable in this mode
Int:13	Bit Transmission Order	Byte: least significant bit (LSB) first - Data: most significant byte first
Int:14	Wake-up Process	Yes
Int:15	Polarization	Omni-directional

6.3.2 Tag to interrogator link

Table 41 — Tag to interrogator link parameters

Ref.	Parameter	Value
Tag:1	Operating Frequency range	433,92 MHz
Tag:1a	Default Operating Frequency	433,92 MHz
Tag:1b	Operating Channels	Not applicable in this mode
Tag:1c	Operating Frequency Accuracy	+/-20 ppm
Tag:1d	Frequency Hop Rate	Not applicable in this mode
Tag:1e	Frequency Hop Sequence	Not applicable in this mode
Tag:2	Occupied Channel Bandwidth	200 kHz
Tag:2a	Minimum Receiver Bandwidth	200 kHz
Tag:3	Transmit Maximum EIRP	5,6 dBm (peak power) or as allowed by local regulations
Tag:4	Transmit Spurious Emissions	
Tag:4a	Transmit Spurious Emissions, In-Band	Not applicable in this mode
Tag:4b	Transmit Spurious Emissions, Out of Band	The tag shall transmit in conformance with spurious emissions requirements defined by the country's regulatory authority within which the system is operated.
Tag:5	Transmit Spectrum Mask	Not applicable in this mode
Tag:6	Timing	
Tag:6a	Transmit to Receive Turn Around Time	1 ms
Tag:6b	Receive to Transmit Turn Around Time	1 ms
Tag:6c	Transmit Power On Ramp	1 ms
Tag:6d	Transmit Power Down Ramp	1 ms
Tag:7	Modulation	Frequency Shift Keying (FSK)
Tag:7a	Spreading Sequence	Not applicable in this mode
Tag:7b	Chip Rate	Not applicable in this mode
Tag:7c	Chip Rate Accuracy	Not applicable in this mode
Tag:7d	On-Off Ratio	Not applicable in this mode

Table 41 (continued)

Ref.	Parameter	Value
Tag:7e	Sub-carrier Frequency	Not applicable in this mode
Tag:7f	Sub-carrier Frequency Accuracy	Not applicable in this mode
Tag:7g	Sub-carrier Modulation	Not applicable in this mode
Tag:7h	Duty Cycle	Not applicable in this mode
Tag:7i	FM deviation	±35 kHz
Tag:8	Data Coding	Manchester, 36 µs bit period. Logic one: 18 µs low followed by 18 µs high, Logic zero: 18 µs high followed by 18 µs low
Tag:9	Bit Rate	27,7 kbit/s
Tag:9a	Bit Rate Accuracy	200 ppm
Tag:10	Tag Transmit Modulation Accuracy	Not applicable in this mode
Tag:11	Preamble	
Tag:11a	Preamble Length	Twenty one (21) bits
Tag:11b	Preamble Waveform	Square wave as defined in Tag:11c
Tag:11c	Bit Sync Sequence	20 cycles of 30 µs high, 30 µs low, followed by one cycle 42 µs high, 54 µs low
Tag:11d	Frame Sync Sequence	20 cycles of 30 µs high, 30 µs low, followed by one cycle 42 µs high, 54 µs low
Tag:12	Scrambling	Not applicable in this mode
Tag:13	Bit Transmission Order	Byte: least significant bit (LSB) first Data: most significant byte first
Tag:14	(Reserved by committee)	
Tag:15	Polarization	Omni-directional
Tag:16	Minimum Tag Receiver Bandwidth	200 kHz

6.3.3 Protocol parameters

Table 42 — Protocol parameters

Ref.	Parameter Name	Description
P:1	Who talks first	Reader-Talks-First (RTF)
P:2	Tag addressing capability	Yes
P:3	Tag UID	Yes
P:3a	UID Length	32 bit
P:3b	UID Format	binary
P:4	Read size	1 to 46 bytes,
P:5	Write Size	1 to 46 bytes,
P:6	Read Transaction Time	27,7 kbit/s
P:7	Write Transaction Time	27,7 kbit/s
P:8	Error detection	CCITT 16
P:9	Error correction	None
P:10	Memory size	1 byte, 128 kbytes
P:11	Command structure and extensibility	8 bits for command

6.3.4 Anti-collision parameters

Table 43 — Anti-collision parameters

Ref.	Parameter Name	Description
A:1	Type	Probabilistic
A:2	Linearity (for N tags)	Probabilistic: $0,065 \cdot N$ seconds for $1 \leq N \leq 3000$
A:3	Tag inventory capacity	Probabilistic: 3000

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